

Candidate Partial Result: A Karhunen–Loeve Basis from the Separating Dual

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Status. Superseded candidate partial result. This packet addresses the first of two questions in Remark 2.26 of Cioica–Licht–Cox–Veraar [1]. It is retained as provenance for the density/basis argument and is superseded by the full candidate packet

`solutions/full/1804.08947_karhunen_loeve_pathwise_expansion/`.

The stronger pathwise expansion is proved in that later packet, not here.

Source Question

The source paper proves a Karhunen–Loeve type expansion for Gaussian random variables in separable r -Banach spaces with separating dual, using Byczkowski’s RKHS theorem [2]. The relevant source passage is Remark 2.26 on page 14:

of [17, Theorem 1].

□

Remark 2.26. It is not immediately clear from [17] whether in the proof of Proposition 2.25 one may choose the orthonormal basis $(h_n)_{n \in \mathbb{N}}$ out of E^* . It is also unclear whether it is possible to choose $(h_n)_{n \in \mathbb{N}}$ in such a way that $\sum_{n=1}^{\infty} h_n(X) x_n = X$ $\mathbb{P}$ -a.s.

Now we explain how to use Proposition 2.25 prove the generalizations of Theo

In words, the source asks whether, in Proposition 2.25, the orthonormal basis (h_n) can be chosen from E^* , and whether it can be chosen so that $\sum_{n=1}^{\infty} h_n(X) x_n = X$ almost surely.

Definitions

Let E be a separable r -Banach space, $0 < r \leq 1$, and let μ be the law of an E -valued Gaussian random variable X . Following Byczkowski’s notation as recalled before Proposition 2.25 in the source paper, let $E^2(\mu)$ denote the real Hilbert space of square-integrable μ -a.e. additive measurable functionals on E , modulo equality μ -a.e., with inner product

$$\langle f, g \rangle_{E^2(\mu)} = \int_E f(x) g(x) d\mu(x).$$

Each $x^* \in E^*$ defines an element of $E^2(\mu)$ by restriction to the support of μ .

Proof Intuition

The separating dual already generates the Borel structure of E . Thus any measurable additive Gaussian functional $g \in E^2(\mu)$ is a function of the random variables $x^*(X)$, $x^* \in E^*$. If g were orthogonal to all continuous linear functionals, joint Gaussianity would make $g(X)$ independent of every finite family of those coordinates, hence independent of the whole sigma-field generated by them. A random variable which is measurable with respect to a sigma-field and independent of it must be constant; since the functional is centered Gaussian, that constant is zero.

Theorem 1. *Let E be a separable r -Banach space whose dual E^* separates the points of E . Let X be an E -valued Gaussian random variable with law μ . Then the canonical image of E^* is dense in $E^2(\mu)$. Consequently, $E^2(\mu)$ admits an orthonormal basis consisting of elements of E^* .*

Proof. Let D be the closed subspace of $E^2(\mu)$ generated by the canonical image of E^* . We prove that $D = E^2(\mu)$.

Take $g \in E^2(\mu)$ orthogonal to D . For every finite family $x_1^*, \dots, x_m^* \in E^*$, the random vector

$$(g(X), x_1^*(X), \dots, x_m^*(X))$$

is jointly Gaussian. Indeed, every real linear combination of its coordinates is again represented by an element of $E^2(\mu)$, and elements of $E^2(\mu)$ are Gaussian in Byczkowski's construction. Moreover,

$$\mathbb{E}[g(X)x_j^*(X)] = \langle g, x_j^* \rangle_{E^2(\mu)} = 0 \quad (j = 1, \dots, m).$$

For jointly Gaussian random variables, zero covariance is independence, so $g(X)$ is independent of each finite vector $(x_1^*(X), \dots, x_m^*(X))$.

Since E is separable and E^* separates points, the source paper recalls that $\mathcal{B}(E) = \sigma(x^* : x^* \in E^*)$, and there is a countable subset of E^* which separates points. Hence $g(X)$, being Borel measurable as a function of X , is measurable with respect to

$$\sigma\{x^*(X) : x^* \in E^*\}.$$

The finite-dimensional independence above implies that $g(X)$ is independent of this sigma-field. Therefore $g(X)$ is independent of itself, and so it is almost surely constant. Elements of $E^2(\mu)$ are centered Gaussian functionals, hence this constant is 0. Thus $g = 0$ in $E^2(\mu)$, and $D^\perp = \{0\}$.

Therefore $D = E^2(\mu)$. Finally, $E^2(\mu)$ is separable because it is a closed subspace of $L^2(E, \mathcal{B}(E), \mu)$ over a countably generated Borel space. Choose a countable dense subset of the canonical image of E^* and apply Gram–Schmidt. Each resulting basis vector is a finite real linear combination of elements of E^* , and therefore itself belongs to E^* . $\square$

Verification Notes

The proof gives a positive answer only to the first sentence of Remark 2.26: the orthonormal basis in $E^2(\mu)$ may be selected from E^* . It does not construct coefficient vectors $x_n \in E$, and it does not prove the pathwise identity $\sum_n h_n(X)x_n = X$ almost surely. The later full candidate packet supersedes this limitation.

No computational check is relevant. A bounded literature check was performed using the run indexes and exact web searches for the remark's wording, “Karhunen–Loeve”, “Byczkowski”, and “orthonormal basis out of E^* ”; no prior packet or direct later answer was found.

Human Review Recommendation

Review the density argument at the point where measurability with respect to $\sigma(E^*)$ and joint Gaussianity of $g(X)$ with the dual coordinates are used. The intended scope is deliberately narrow; reviewers should not read the packet as resolving the almost-sure series representation question.

References

- [1] P. A. Cioica-Licht, S. G. Cox, and M. C. Veraar, *Stochastic integration in quasi-Banach spaces*, arXiv:1804.08947.
- [2] T. Byczkowski, *RKHS for Gaussian measures on metric vector spaces*, Bull. Polish Acad. Sci. Math. 35 (1–2), 93–103, 1987.